

HACKATHON PROBLEM STATEMENTS

Build. Break. Defend.

Two tracks: Game Development & Behavioral Security

Terminal Mafia (36 hrs) | **BioPrint: Behavioral Auth (36 hrs)**

Event 1 | Terminal Mafia

A local-hosted, multiplayer, terminal-based social deduction game

Duration: 36 hours

Hosting: Local device / LAN / hotspot

Theme: Terminal Mafia

Problem Statement

Build a multiplayer, terminal-based (CLI) social deduction game in the spirit of Mafia / Among Us. Players are assigned hidden roles (e.g., Mafia vs. Villagers, or a Betrayer among Cooperators), and the game must support a discussion/action phase, a voting or elimination mechanic, and distinct information visible only to each player based on their role.

The game must be playable entirely within the terminal (no GUI) and hosted locally: either on a single machine using multiple terminal windows, or across devices connected over the same local network or hotspot. No cloud deployment is required.

Core Requirements

- Supports 4+ players with hidden, role-based information (each player only sees what their role permits)
- A structured game loop: role assignment → phases (e.g., night/day, discuss/vote) → elimination → win condition
- Fully playable via terminal input/output: text-based only, no GUI frameworks
- Reasonably graceful handling of a player disconnecting or submitting invalid input mid-game
- A clear, unambiguous win condition for each side (e.g., Mafia eliminated / Mafia outnumbers Villagers)

Stretch Goals (optional, bonus credit)

- AI-controlled bot players to fill out a lobby when fewer humans are available
- Spectator mode for eliminated players
- Additional special roles (e.g., Detective, Doctor, Double Agent) beyond the basic two sides
- Match history / replay of role reveals and voting patterns at game end

Deliverables

Item	Description	Non-Submission Penalty
Git Repository	The team must share the Git Repository containing all the source code for the game along with a valid commit history and a ReadMe file	Disqualification
Executable File	Each Team must submit an executable file that can be directly loaded	-20 points
Demonstration Video	Each Team must submit a demonstration 3–5-minute demonstration video of their game played by multiple individuals	-10 points

Judging Criteria

Criterion	What judges look for	Score
Gameplay & Fun Factor	Is it genuinely engaging to play (tension, bluffing, pacing), not just functionally complete?	30 points
Execution Quality	Polish, stability, and responsiveness during the live demo. A smaller, well-executed game beats an ambitious, buggy one.	30 points
Creativity & Vision	Original twists on the Mafia formula, interesting role design, or clever use of the terminal medium.	25 points
Technical Craft	Code structure, handling of hidden per-player state, and quality of the terminal UI/UX under text constraints.	15 points

Demo format: Each team runs a live 3–5 minute play session with real players (teammates or judges), showing at least one full phase cycle (discussion → vote → elimination).

Event 2 | BioPrint: Behavior-Based Login Security

A Chrome extension or website login system that authenticates people by how they behave, not just what they type

Duration: 36 hours

Hosting: Local device, browser extension, or any free-tier web host

Theme: Passwordless-proof identity through behavioral biometrics

Problem Statement

Passwords can be guessed, phished, or leaked, but the way a person behaves while using a page is far harder to imitate. Build a Chrome extension or a website login page that authenticates users using behavioral biometrics. Which behavioral signals to capture, and how to capture them, is entirely up to each team.

The core challenge is this: even if an attacker has the correct password, the system should still be able to recognize that the person behind the keyboard is not the genuine account owner and block the login purely by comparing live behavior against the user's established pattern. This detection must work without falling back on OTPs or any other secondary-verification step, and it should equally be able to flag non-human traffic, such as scripted or bot-driven login attempts, that lacks any believable human behavioral signature.

During enrollment, teams may design any onboarding flow they choose, of any format and length, to capture each user's baseline behavior. From that point on, the system should be able to tell a genuine returning user apart from an impostor trying to mimic them, based on pattern alone. Teams have complete freedom to choose their own statistical, heuristic, or machine-learning approach to building each user's unique behavioral fingerprint, so long as that fingerprint is independent of the password itself.

Core Requirements

- An enrollment flow, designed entirely by the team, that collects each user's baseline behavioral data
- A behavioral fingerprinting engine that models each user's unique behavioral patterns, independent of their password
- A live authentication check that compares a login attempt's real-time behavior against the enrolled profile and can block access on a mismatch, with no OTP or secondary-verification fallback
- Detection of non-human / automated login attempts (bots, scripts, replayed input) as a distinct fraud signal
- A working demo login page or Chrome extension showing successful logins for genuine users and blocked logins for mimicked or bot attempts

Stretch Goals (optional, bonus credit)

- Adaptive profiles that gently update as a genuine user's behavior naturally drifts over time
- A confidence score or live dashboard showing why an attempt was accepted or flagged
- Support for multiple input modalities (touchpad, external mouse, mobile touch) within one profile
- Explainability: a human-readable summary of which behavioral signals triggered a block

Deliverables

Item	Description	Non-Submission Penalty
Git Repository	The team must share the Git Repository containing all the source code for the extension along with a valid commit history and a ReadMe file	Disqualification
Executable File	Each Team must submit an executable file that can be directly loaded onto Google Chrome or have the extension embedded in the website	-20 points

Item	Description	Non-Submission Penalty
Report	A 1–2-page report detailing the working of your idea and the core logic behind your authentication algorithm	-10 points
Dummy Login Page	A dummy login page in which this can be tested on, you must send the code for the page and the setup instructions to run it in Local Host or have the website hosted in any public hosting service	-20 points

Judging Criteria

Criterion	What judges look for	Score
Creativity	Originality of the behavioral signals chosen and how imaginatively they're combined into a fingerprint.	10 points
Reliability	Consistency of correctly accepting genuine users and rejecting impostors across repeated attempts.	25 points
Ease of Use	How natural and unobtrusive the enrollment and login experience feels to a real user.	10 points
Customer Satisfaction	Overall confidence and comfort a user would feel trusting this system with their account.	10 points
Novelty in Algorithms	Sophistication and originality of the statistical or ML approach used to model and match behavior.	15 points
Latency	Speed of the authentication decision; the system should feel instant, not like a background batch job.	10 points
Overall Innovation	How compelling and demo-ready the solution is as a genuine rethink of login security.	20 points

Demo format: Each team runs a live login demo showing (a) a genuine user enrolling and logging in successfully, and (b) a teammate attempting to log in with the correct password but the wrong behavior, or via a scripted/bot attempt, being detected and blocked.